1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist identified a tension between local immune organization (non-invasive immune readouts like TLS/TILs) and molecular determinants of tumor behavior (MSH6/MMR status and GREM1-driven stromal/BMP signaling): why do immune-rich bladder tumors sometimes fail to respond to immune checkpoint therapy or still progress? 2) Tier 1 paths used: I built on the MSH6 (MMR/prognosis/immunogenicity) path because MSH6 loss elevates neoantigen load and can drive ICI sensitivity, and the GREM1 (BMP/stromal, metastasis) path because GREM1-mediated stromal remodeling can create immune-excluding microenvironments; these two mechanisms directly explain when immune structures visible on H&E may or may not translate into effective anti-tumor immunity. 3) Tier 2 evidence supporting the claim: CTCs and urine-DNA methylation panels provide orthogonal, non-invasive readouts of systemic dissemination and tumor epigenetic state that could stratify tumors with discordant imaging vs molecular signals; HDL-C and GARS1 suggest systemic metabolic/immune-modulatory axes that could modulate stromal or immune function, reinforcing the plausibility of a stromal/immunometabolic barrier. 4) Why therapy_response was chosen as the endpoint: MSH6 implicates MMR-driven immunogenicity and the central question is why immune-infiltrated tumors do or do not benefit from checkpoint blockade—thus predicting ICI/immune therapy response is the most mechanistically relevant and clinically actionable endpoint. 5) Deliberate exclusions: I limited the hypothesis to one imaging feature plus two anchor genes to preserve mechanistic clarity—therefore urine-DNA methylation (a Tier1 biomarker category) and other Tier2 items (CTCs, HDL-C, GARS1) were moved to secondary_modifiers rather than included in the core claim; exploratory/cross-disease paths (PSA, SARIFA, CSF-AD biomarkers) were excluded due to weak bladder-cancer relevance?

Working Hypothesis

Refined hypothesis: In bladder cancer, TLS-like lymphoid aggregates predict responsiveness to immune checkpoint inhibitors (ICIs) only in tumors with low BMP/GREM1 signaling and with MSH6 deficiency (which elevates neoantigen burden). High GREM1-driven BMP signaling induces stromal remodeling that creates an immune-excluding microenvironment, rendering TLS presence insufficient to predict ICI benefit. Therefore, a composite pre-treatment biomarker that combines TLS assessment with MSH6 status and GREM1/BMP activity will stratify patients more accurately for ICI response and identify those who may require concurrent anti-stromal or BMP-targeted therapy to unleash TLS-mediated anti-tumor immunity. To test this, we will: (1) perform retrospective analyses of ICI-treated bladder cancer cohorts with data on TLS, MSH6 status, and GREM1/BMP signaling; (2) prospectively validate a standardized TLS scoring and molecular readout in an independent cohort, assessing MSH6 and GREM1/BMP activity; (3) conduct mechanistic ex vivo and in vivo experiments to determine whether BMP inhibition or GREM1 suppression rescues TLS-associated cytotoxicity; and (4) evaluate whether incorporating urine DNA methylation (as a secondary modifier) and other Tier2 readouts refines predictive accuracy without diluting the core signal.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, GREM1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6/MMR status and GREM1-driven stromal/BMP signaling): why do immune-rich bladder tumors sometimes fail to respond to immune checkpoint therapy or still progress? 2) Tier 1 paths used: I built on the MSH6 (MMR/prognosis/immunogenicity) path because MSH6 loss elevates neoantigen load and can drive ICI sensitivity, and the GREM1 (BMP/stromal, metastasis) path because GREM1-mediated stromal remodeling can create immune-excluding microenvironments; these two mechanisms directly explain when immune structures visible on H&E may or may not translate into effective anti-tumor immunity.

GREM1: GREM1-driven stromal/BMP signaling): why do immune-rich bladder tumors sometimes fail to respond to immune checkpoint therapy or still progress? 2) Tier 1 paths used: I built on the MSH6 (MMR/prognosis/immunogenicity) path because MSH6 loss elevates neoantigen load

and can drive ICI sensitivity, and the GREM1 (BMP/stromal, metastasis) path because GREM1-mediated stromal remodeling can create immune-excluding microenvironments; these two mechanisms directly explain when immune structures visible on H&E may or may not translate into effective anti-tumor immunity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	urine-DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- urine-DNA methylation biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.